

How the BITCOLLAGE Cutting Tool Platform Works

A simple guide for CNC machining MSMEs

AI-Driven Tool Selection & Procurement Platform — Understanding the Flow

1. What this platform is

The BITCOLLAGE Cutting Tool Platform is a **digital procurement assistant** built specifically for CNC machining MSMEs. Today, deciding *which cutting tool to buy and at what parameters to run it* depends on the experience of one or two senior operators, distributor recommendations, and a fair amount of guesswork.

This platform replaces guesswork with **engineering-driven AI recommendations** — and connects those recommendations directly to your ERP, your shopfloor, and your authorised distributors. The result: the right tool, the right parameters, ordered automatically when production needs it.

2. Who uses it

Role	What they get from the platform
Plant Owner / MSME Head	Visibility into tool spend, downtime saved, productivity gains
Production / Tool Room Engineer	Instant tool recommendations with cutting parameters
Procurement Team	Auto-triggered purchase plans from real production data
Operators	Pre-validated tools that won't fail mid-job
Distributors	Continued participation through a transparent, fair price-comparison layer

3. The end-to-end flow — at a glance



4. Screen-by-screen guide

4.1 Landing Page (Pitch Overview)

URL: /

The landing page is the public-facing introduction to the platform. It explains the problem, the four core capabilities, the way the platform fits into existing workflows, and the impact MSMEs can expect.

What you'll see:

- A live preview of an AI recommendation (CNMG insert with cutting parameters)
- The problem MSMEs face today (6 pain points)
- The four solution pillars (Recommendation Engine, ERP Integration, Distributor Layer, Learning Engine)
- A side-by-side comparison: typical marketplace vs. BITCOLLAGE's engineering-first approach
- The 7-step closed-loop flow
- Forecasted impact numbers

Call-to-action: "Launch Interactive Prototype" — takes the visitor inside the platform.

4.2 Dashboard

URL: /app/dashboard

The first screen after entering the platform. Designed for the plant owner or production manager who wants a **30-second daily snapshot**.

What it shows:

- **4 KPI tiles** — Active Work Orders, Tools Procured This Month, Avg Tool Life vs Target, Downtime Hours Saved
- **Recent Recommendations table** — what the AI has been advising today (with confidence scores)
- **Alerts panel** — issues the learning engine wants you to know about (e.g., "Insert grade XYZ failing 30% above expected")
- **Quick Actions** — jump directly into Tool Recommendation, ERP, or Analytics

Why this exists: Decision-makers don't have time to dig — the dashboard surfaces what's important without them asking.

4.3 Tool Recommendation Engine — *The Core Differentiator*

URL: /app/tool-recommendation

This is the heart of the platform. A production engineer who needs to start a new job comes here.

How it works — step by step:

Step 1 — Tell the engine your inputs (left panel):

- Workpiece material (e.g. EN24 alloy steel)
- Operation type (Turning / Milling / Drilling)

- Machine (e.g. Mazak QTN-200MY)
- Target surface finish (Ra in μm)
- Tool diameter

Step 2 — Hit *Get Recommendation*: The platform analyses your inputs against:

- A library of tool grades and geometries
- Your machine's capability
- Past performance data from your own shopfloor
- Material/operation/tool compatibility rules

Step 3 — Get the optimal recommendation (right panel):

- The **specific tool** to use (brand, code, type — e.g. *Sandvik CNMG 120408-PM 4325*)
- **Cutting parameters** — Vc (cutting speed), f (feed rate), ap (depth of cut)
- **Estimated tool life** in parts
- A **confidence ring** showing how certain the AI is
- **Unit price** with a note on where the best price was found

Step 4 — Compare or order:

- See alternative recommendations (other brands, slightly different geometries)
- Click *Add to Procurement* to send the tool into your purchase plan

Why this is the core differentiator: Other platforms make you search a catalog. This platform tells you *exactly* what to buy and *exactly* how to run it — based on your actual job conditions.

4.4 ERP / Shopfloor Integration

URL: `/app/erp-integration`

This screen demonstrates the most powerful idea in the platform: **procurement starts from real production need, not a manual request.**

What it shows:

- A **live connection status** banner (currently connected to SAP B1 / Tally with last sync time)
- A **list of all active work orders**, auto-fetched from your ERP
- Quick stats — total work orders, BOM items, tools tracked

When you click any work order, the right panel shows:

- WO number, component name, quantity, material, due date, status
- The **operations chain** (e.g. Facing → Turning → Drilling → Threading)
- An **auto-procurement banner** — the platform has already identified which tools this job needs and has a procurement plan ready for your approval

- The **BOM** (Bill of Materials), auto-fetched
- **Tool consumption history** — how many tools were consumed on similar past jobs, and how many failed

Why this is powerful: Today, procurement happens when someone walks to the store and says "we're out of inserts." With this platform, the system *already knows* you'll need 8 specific inserts for next week's work orders — and has prices, availability, and a draft PO waiting.

4.5 Performance Feedback & Analytics

URL: `/app/analytics`

This is where the platform proves its value over time, and where the AI gets smarter.

4 KPI tiles at the top:

- **Avg Tool Life** — how many parts each tool is actually producing
- **Failure Rate** — how often tools fail before expected life
- **Rejection Rate** — how often parts get rejected
- **Feedback Points** — how many data points the AI learned from this month

Charts:

- **Tool Life Trend (12 weeks)** — actual vs target. A rising line = the AI's recommendations are working.
- **Failures by Tool Grade** — which grades fail most. Helps you spot a bad batch or a wrong recommendation.
- **Rejection Reasons** — breakdown of *why* parts get rejected (surface finish, dimensional, burr, etc.)

Learning Insights panel: This is plain-English commentary from the AI:

- *"Recommendations updated based on 142 feedback points last month."*
- *"Ceramic RNGN failing 30% above expected on Inconel 718 — switched recommendation to CBN insert."*
- *"Tool life on CNMG 120408 PM 4325 consistently 10% above target — increased recommended Vc from 260 → 280 m/min."*

Tool Utilization table: every active tool, expected vs actual life, and a color-coded status.

Why this exists: The pitch promises that the AI gets better over time. This screen makes that visible — both as numbers and as readable insights.

5. What happens "behind the scenes"

Engineering layer

When you ask for a tool recommendation, the engine isn't running a simple lookup. It uses:

- **Material machinability databases** (hardness, chip behaviour, thermal conductivity)
- **Tool geometry libraries** (insert grades, edge preparations, coatings)
- **Machine capability data** (max RPM, available coolant pressure, rigidity class)
- **Operation-specific rules** (turning vs milling vs drilling parameter ranges)

AI / Learning layer

Over time, the platform learns from *your specific shopfloor*:

- Which tools actually last longer than the manufacturer claims (in your conditions)
- Which combinations cause chatter or burr on your particular machines
- Which distributors deliver on time vs. delay shipments
- Which operators consistently get longer tool life — and what they do differently

ERP / Workflow layer

The platform doesn't replace your ERP — it **augments** it:

- Reads work orders, BOMs, stock levels
- Writes back tool consumption, purchase orders, performance data
- Keeps your existing ERP (SAP B1, Tally, custom systems) as the single source of truth

Distributor layer

Authorised distributors plug into the platform via simple integrations:

- Live stock availability
- Real-time pricing
- Lead-time commitments
- Quality assurance and after-sales feedback

No distributor is locked out. The MSME always sees options and picks the best fit.

6. What the MSME actually gains

Before	After
"We always use whatever the distributor recommended last time"	"The AI tells me the best tool for <i>this</i> material, <i>this</i> machine, <i>this</i> operation"
Tool failure mid-shift → 2 hours of downtime	Pre-validated tools → 25–40% improvement in tool selection accuracy
Reorder when the store guy says we're out	Reorder <i>before</i> we're out, based on actual production demand
Distributor relationships are opaque on pricing	Transparent, side-by-side comparison every time
Every job is a new guess	Every job adds to a learning database that benefits the next job

7. Why this works for MSMEs specifically

Big factories have engineering teams, CAM software, and tool management systems. MSMEs don't. This platform brings the same level of decision intelligence to a 5-machine shop that a 500-machine plant has — **at a SaaS price, without disrupting the existing distributor ecosystem.**

It's built around three principles:

1. **Embedded, not external** — works inside your ERP and shopfloor, not as a separate tool you have to remember to open
2. **Engineering-first, not marketplace-first** — recommends what's *right*, not just what's cheapest
3. **Adaptive, not static** — gets smarter the more you use it

8. Where to start

1. **Connect your ERP** (one-time setup)
2. **Import your existing tool list** (so the platform knows what you already own)
3. **Start with one machine or one product family** — see the recommendations, log the results
4. **Roll out to the rest of the shopfloor** as you build confidence

Within 3 months, the learning engine will have enough shopfloor-specific data to give you measurable improvements in tool life and downtime.

9. Quick reference — Platform navigation

Section	What it's for	Who uses it most
Landing Page	First-time visitors learning what the platform does	Prospective customers, executives
Dashboard	Daily snapshot of operations	Plant owners, production heads
Tool Recommendation	Get an optimal tool + parameters for a specific job	Tool room engineers, operators
ERP Integration	See work orders, BOM, auto-procurement plans	Procurement, production planning
Analytics	Tool performance, failures, learning insights	Quality, engineering, management

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This document explains the platform flow for a general business audience. For a deeper technical specification or integration guide, please contact BITCOLLAGE directly.
